



The University of Edinburgh



College of Science and Engineering

Mathematics 3 Honours
MATH10003 Financial Mathematics

Friday, 16th December 2016

14:30 – 16:30

Chairman of Examiners – Dr A Olde Daalhuis

External Examiner – Professor J McColl

Answer both questions

Calculators and other electronic aids

A scientific calculator is permitted in this examination.

It must not be a graphical calculator.

It must not be able to communicate with any other devices.

This examination will be marked anonymously.

- (1) (a) Let $(W_t)_{t \geq 0}$ be a standard Brownian motion. Fix $r \in (0, \infty)$. Prove that the process $(B_t)_{t \geq 0}$ defined as $B_t := W_{t+r} - W_r$ is a Brownian motion.

[7 marks]

- (b) The following stochastic differential equation is known in finance as the Vasicek model

$$dr_t = \lambda(\mu - r_t)dt + \sigma dW_t,$$

where $(W_t)_{t \geq 0}$ is a standard Brownian motion, $\lambda, \mu, \sigma \in (0, +\infty)$ and r_0 is the initial value.

Prove, by applying Itô's formula to $Y_t := e^{\lambda t} r_t$, that

$$r_t = \mu + (r_0 - \mu)e^{-\lambda t} + \sigma \int_0^t e^{\lambda(s-t)} dW_s \quad \text{for every } t \geq 0.$$

[6 marks]

- (c) Let $T > 0$, $t \in [0, T]$ and take the function $\mu : [0, T] \times \mathbb{R} \rightarrow (0, \infty)$ and $\sigma : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$. Let $(X_s)_{s \in [t, T]}$ be the Itô process defined as

$$\begin{aligned} dX_s &= \mu(s, X_s)ds + \sigma(s, X_s)dW_s, \quad \text{for } s \in [t, T] \\ X_s &= x. \end{aligned}$$

Assume that $f \in C^{1,2}([0, T] \times \mathbb{R}, \mathbb{R})$ is a solution to the following boundary value problem PDE for $(t, x) \in [0, T] \times \mathbb{R}$

$$\partial_t f(t, x) + \mu(t, x)\partial_x f(t, x) + \frac{1}{2}\sigma^2(t, x)\partial_{xx} f(t, x) - rf(t, x) = 0,$$

and boundary condition $f(T, x) = \Phi(x)$ for $x \in \mathbb{R}$ for a function $\Phi \in C^1(\mathbb{R}, \mathbb{R})$.

Assume that the process $\left(e^{-rs}\sigma(s, X_s)\partial_x f(s, X_s)\right)_{s \in [t, T]} \in L^2([t, T])$.

Apply Itô's formula to the process $e^{-sr}f(s, X_s)$ for $s \in [t, T]$ to prove that for any $(t, x) \in [0, T] \times \mathbb{R}$ the value $f(t, x)$ can be expressed by

$$f(t, x) = e^{-r(T-t)}\mathbb{E}[\Phi(X_T)].$$

[12 marks]

- (2) (a) An analyst is investigating the extent to which the price of a stock at the beginning and end of a time interval can be used to provide information about its price during the interval. The model used is

$$S_t = S_0 + \mu t + \sigma W_t,$$

where $\{W_t\}_{t \geq 0}$ is a standard Brownian motion. The analyst wishes to investigate the difference between S_t and the price $\hat{S}_t$, given by:

$$\hat{S}_t = \frac{(T-t)S_0 + tS_T}{T}, \quad \text{for every } t \in [0, T].$$

- (i) Calculate the following *five* quantities: $\mathbb{E}[S_t - S_0]$, $\text{Var}[S_t - S_0]$, $\mathbb{E}[S_T - S_t]$, $\text{Var}[S_T - S_t]$ and $\text{Cov}(S_t - S_0, S_T - S_t)$ for $0 \leq t \leq T$. [5 marks]
 - (ii) Calculate the expectation of $S_t - \hat{S}_t$. [2 marks]
 - (iii) Calculate the variance of $S_t - \hat{S}_t$. [3 marks]
 - (iv) Find the value of $t \in [0, T]$ where the variance of $S_t - \hat{S}_t$ is greatest. [2 marks]
- (b) Suppose that $T \in (0, \infty)$, that $(W_t)_{t \in [0, T]}$ is a standard Brownian motion and that the price of a stock at time t is given by

$$S_t = S_0 e^{(r - \frac{\sigma^2}{2})t + \sigma W_t},$$

for $t \in [0, T]$, where $S_0 > 0$ is the current price of the stock, $r > 0$ is the risk-free interest rate (continuous compounding) and $\sigma \in (0, \infty)$.

Consider the European-style super-share option $h(S_T)$ written on the non-dividend paying stock with the following pay-off at maturity time T

$$h(S_T) := \begin{cases} \frac{S_T}{K_1} & , \text{ if } S_T \in [K_1, K_2] \\ 0 & , \text{ otherwise} \end{cases}$$

where $0 < K_1 < K_2 < \infty$.

Prove that the price of the option $h(S_T)$ (its discounted expected payoff) written on the stock S is given (by the expression on the right)

$$\Pi(0; h(S_T)) := e^{-rT} \mathbb{E}[h(S_T)] = \frac{S_0}{K_1} (\Phi(d_1) - \Phi(d_2))$$

where Φ is the cumulative probability distribution function of the standard normal distribution and

$$d_1 = \frac{\ln\left(\frac{S_0}{K_1}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, \quad \text{and} \quad d_2 = \frac{\ln\left(\frac{S_0}{K_2}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}.$$

[13 marks]

